ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATIONS: 2024-25

B.Tech.

Semester: II

SUBJECT CODE: BAS-202

Subject Name: Mathematics-II

Time : 3 HOURS ***Maximum Marks: 70***

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS
ATTEMPT ALL THE SECTIONS:

SECTION- A

NOTE: Attempt all parts of the following question. (14 Marks)

Q. 1. Answer the following questions: (1x7 = 7 Marks)

Write the iterative formula of Newton Raphson method.

Write the formula of Euler 's method.

Evaluate $\int_0^a \int_0^b (x^2 + y^2) dx dy$

Define Interpolation and Extrapolation.

Find the value of $(1-\nabla)(1+\Delta)$.

Write the formula of correlation coefficients.

Find the value of $\Gamma(1/4)\,\Gamma(3/4)$.

Q. 2. Answer the following questions: **(1x7 = 7Marks)**

Write the formula of central difference operator.

Define leptokurtic.

Write the normal equation of $y = ax^b$.

Write the formula of Trapezoidal rule to find $\int_a^b f(x)\,dx$.

Write the formula of Regression coefficient.

Evaluate $\int_{-1}^{1}\int_{-2}^{2}\int_{-3}^{3} dxdydz$.

Evaluate $\beta\left(\frac{3}{2},\frac{5}{2}\right)$.

SECTION-B

NOTE: Attempt all the questions. **(7x2x2 = 28 Marks)**

Q. 3. Answer the following: **(7x2=14)**

(i) Find by method of Regula –Falsi a root of the equation. $x^3 + x - 1 = 0$ lying between 1and 2.

OR

Prove that (a) $\mu\delta = \frac{1}{2}(\Delta + \nabla)$

(b) $e^x = \left(\frac{\Delta^2}{E}\right)e^x \frac{Ee^x}{\Delta^2 e^x}$

(ii) Solve the equations, by Gauss –Seidel method,

$3x + y + z = 1, x + 3y - z = 11, x - 2y + 4z = 21.$

OR

Prove that $\frac{B(p,q+1)}{q} = \frac{B(p+1,q)}{p} = \frac{B(p,q)}{p+q}$

Q. 4. Answer the following: **(7x2=14)**

(i) Using method of least square , find the curve $y = ax + bx^2$ that best fits the following data:

x	1	2	3	4	5
y	1.8	5.1	8.9	14.1	19.8

OR

Find the Rank correlation coefficient of marks of A and B from the following data:

x:	68	64	75	13	50	64	80	75	40	55	64
y:	62	58	68	30	45	81	60	68	48	50	70

(ii) Apply Newton's divided difference method to obtain an interpolatory polynomial for the following data;

x	1	3	5	7	10
y	13	31	25	37	101

OR

Use Lagrange's interpolation formula to find value of $f(x)$, at x=2.5 from the following table:

x	1	2	3	4
$f(x)$	1	8	27	64

SECTION-C

NOTE: Attempt any two questions. (14x2 = 28 Marks)

Q. 5. Answer one of the following: (14x1=14)

Using Runge-Kutta method of fourth order to solve $\frac{dy}{dx} = x + y^2$ with y (0) = 1 for x = 0.2.

OR

The following table gives the number of accidents that took place in an industry during various days of the week. Use chi-square test if accidents are uniformly distributed over the week at 5% level of significance.

Day	Mon.	Tues.	Wed.	Thu.	Fri.	Sat
No.of accidents	14	18	12	11	15	14

Note: The table value of chi-square $\left(\chi^2\right) = 11.071$ at 5% level of significance with 5 df.

Q. 6. Answer one of the following: (14x1=14)

Evaluate by changing the order of integration $\int_0^1 \int_{y^2}^{2-y} xydxdy$.

OR

Using Green's theorem find the area of the region in the first quadrant bounded by the line $y = x,\ y = \frac{1}{x}$ and $y = \frac{x}{4}$.

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